

Pilot Planning and Testing

Objective

The objective of this pilot phase is to evaluate the effectiveness of the proposed automated Stop and Lock system by conducting a trial in a working environment. The focus is on minimizing risks during installation, gathering performance data, and is accepted by the people who will use it every day.

Choosing a Test Location

The pilot will be conducted at a carefully selected quiet yard but still has the same setup as other yards in terms of layout, vehicle flow, and weather exposure. This will allow the team to monitor performance and troubleshoot issues without causing major disruptions.

The pilot will be restricted to 1–2 docking bays that allows us to test it on a smaller scale and make changes if needed. The system will not be modified during the peak periods between Black Friday and New Year to comply with Amazon's operational stability requirements.

Installation Plan and Testing condition

A step-by-step installation plan will be developed in coordination with facilities management and safety officers to ensure proper bollard placement, green/red light synchronization, and integration with the existing combi lock system. Safety signs and temporary instructions will be provided to help drivers and Yard Marshalls understand how to use the new system.

The system will be tested in varied conditions to assess real-world robustness, including:

- Day and night shifts
- Rain, snow, or extreme weather
- High and low visibility conditions

This will help us to observe how the automated system performs under different environmental and real-world scenario.

Evaluation Criteria

- **Safety:** Does it reduce Yard Marshall exposure to moving vehicles?
- **Efficiency:** Is it faster than manual Stop sign placement?
- **Reliability:** Does it work consistently in all-weather condition?

Fallback Plan

If the automated system fails, a manual switch will be used during the pilot to guarantee safe and uninterrupted operations. Every incidence will be monitored for analysis.

Feedback and Results

Feedback will be gathered from Yard Marshalls and warehouse staff through surveys, and discussions. This feedback will support final recommendations and ensure continuous improvement aligned with Amazon's safety-first culture.